DATA ITEM DESCRIPTION
Title: BUILT-IN-TEST MEASUREMENT SUMMARY (BMS)

Number: DI-SESS-82334
AMSC Number: N10212
DTIC Applicable: No
Preparing Activity: SH
Applicable Forms: N/A

Approved Date: 20201221
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2020-024

Use/relationship: The Built-In-Test (BIT) Measurement Summary (BMS) provides the Government the measurement requirements of the system, subsystem, or equipment, as well as the accuracy of the BIT equipment and sensors that are installed to assure traceability of all measurements through an unbroken chain of comparisons. The summary identifies and validates the adequacy of the selected BIT equipment and sensors. Where the BIT processing software is used to improve the accuracy of Pass or Fail decisions, the details of that algorithm, as well as its effectiveness, are documented in this summary.

a. The BMS Data Item Description (DID) satisfies the requirements of the following instructions:

1. Department of Defense Instruction (DoDI) 5000.02 Operation of the Defense Acquisition System
https://www.esd.whs.mil/DD/

2. Naval Air Systems Command Instruction (NAVAIRINST) 13640.1 Naval Aviation Metrology and Calibration Program
https://myteam.navair.navy.mil/km/71/Directives/Directives_Instructions/

b. This DID contains the format, content, and intended use information for the data deliverable and is applicable to the acquisition of all military systems, subsystems, and equipment.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions shall be as specified in the contract.

2. Format. Contractor format is acceptable. Sample format BMS template is provided.

2.1 The BMS shall be structured as follows:

a. Cover page.

b. Revision status.

c. Introduction.

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d. Table of contents.

e. List of abbreviations, symbols, and acronyms.

f. Summary Data Table. The Summary Data Table shall be a single, in-line spreadsheet or, if the data is split into multiple pages as in the sample BMS template, a Content ID column shall be used to correlate the associated data.

3. Content. The BMS shall detail the measurement requirements of the system, subsystem, or equipment, as well as the accuracy of the BIT equipment and sensors that are installed to assure traceability of all measurements through an unbroken chain of comparisons. Where the BIT processing software is used to improve the accuracy of Pass or Fail decisions, the details of that algorithm, as well as its effectiveness, shall be documented in the summary. The BMS shall include the following content:

3.1 Identify all operational system, subsystem and equipment BIT measurement requirements and show traceability to the discrete component performing the measurement.

3.2 Objective evidence that all supporting BIT equipment and sensors that are installed in the system, subsystem, or equipment are adequate to support the BIT program.

3.3 Objective evidence that the BIT Processing algorithms are adequate to support the BIT program.

3.4 Administrative information. Classified information shall not be listed in the BMS. Classified parameters and information shall be addressed in a classified supplement or appendix.

3.5 Cover page. The cover page shall include descriptive information such as system or program name, contract number, contractor's company name, current revision, date of submittal, Contract Data Requirements List (CDRL) number, etc.

3.6 Revision status. This section shall be included in the BMS. The initial BMS submittal shall specify "original" or "-" on the revision status page(s). Subsequent BMS revision numbers shall be recorded on the cover page.

3.7 Introduction. This section shall contain general information, remarks, or other information about the system that the preparer feels would be beneficial.

3.8 Table of contents. This table shall reflect the contents and page location numbers of each structural part of the BMS identified in 2.1 (above).

3.9 List of abbreviations, symbols, and acronyms. This list shall include all abbreviations, symbols, and acronyms used in the BMS with their meanings. Abbreviations shall be in accordance with The American Society of Mechanical Engineers (ASME) Y14.38,

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Abbreviations and Acronyms for Use on Drawings and Related Documents, where applicable. https://www.asme.org/

3.10 Summary Data Table. This section is an inline presentation of system, subsystem, etc., the required measurement, failure mode information, the measurement devices employed, the BIT processing components, and BIT algorithm information used in the implementation of BIT. The summary data shall include the following:

3.10.1 Summary Data Table Header. This shall precede the spreadsheet data and shall list the following:

3.10.1.1 Program Name.

3.10.1.2 System Model and Type Designation.

3.10.1.3 Date. (If Summary Data Table date differs from overall BMS date.)

3.10.1.4 Version Number. (If a Summary Data Table version number is assigned that differs from overall BMS revision.)

3.10.2 Content Identification (ID). This column is sequentially numbered starting at number "1" and shall provide a method of reference to uniquely identify the measurement or test that is being discussed.

3.10.3 Operational Equipment. These columns shall list the system, subsystem, Weapon Replaceable Assembly (WRA), or Line Replaceable Unit (LRU) being tested by BIT, along with the Reference Designation (REF DES) of the item being tested.

3.10.4 Measurement. These columns shall list the BIT measurement requirements. When BIT is performing a qualitative check (i.e., no quantitative measurement is taken), "QUAL" shall be in the "Value Range or Value with Tolerance" column.

3.10.5 Failure Modes, Effects, and Criticality Analysis (FMECA). These columns shall list the FMECA Failure Mode(s) that the measurement or test is attempting to identify. These columns shall also identify whether the item has been classified as a Critical Safety Item (CSI). A Critical Safety Item (CSI) is a part, assembly or support equipment whose failure could cause loss of life, permanent disability or major injury, loss of a system or significant equipment damage.

3.10.6 Measurement Device. These columns shall uniquely identify the device performing the quantitative measurement and its measurement range and tolerance (accuracy). For the purposes of this deliverable, Test Accuracy Ratio (TAR) shall be defined as the ratio of the Measurement accuracy tolerance to the Measurement Device tolerance and shall be computed as follows:

TAR = Measurement_Tolerance / Measurement_Device_Tolerance

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In the case of qualitative checks, these columns shall not be populated.

3.10.7 BIT Processing Component (Level 1). These columns shall identify the first logical device in the BIT hardware chain, which analyzes the measured value and provides status up the chain to the BIT Processor in order to facilitate the BIT determination of whether or not the Operational Equipment has passed or failed a given test. For qualitative tests, this component shall be the lowest level component that is performing the qualitative check.

3.10.8 BIT Processing Component (Level 2, 3, etc.). These columns shall identify the next logical device in the BIT hardware chain, which analyzes the measured value and provides status up the chain to the BIT Processor in order to facilitate the BIT determination of whether or not the Operational Equipment has passed or failed a given test. For qualitative tests, this component shall be the next level component that is performing the qualitative check. (Note: There may be more than 2 levels of logical components between the Measurement Device and the BIT Processor. In these cases, additional columns (Level 2, 3, etc.) shall be added to address subsequent levels and show an unbroken chain of devices or components from the measurement or test performed all the way to the BIT Processor.)

3.10.9 BIT Processing Algorithm. This section shall uniquely identify the test being performed, the BIT states and modes in which the test is performed (an individual test may be performed in multiple states and modes), if any filtering is applied, and an explanation of the filtering and post processing being performed, to include discussions related to how the algorithm improves pass or fail determinations and the algorithm's effectivity. Lengthy explanations and discussions that cannot fit into the "Processing Comments" column shall be included in an appendix to the BMS. When these explanations and discussion are in an appendix, the "Processing Comments" column shall uniquely reference the location of this information.

<Sample BMS Template begins on following page>

End of DI-SESS-82334

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Sample Built-in-Test Measurement Summary (BMS) Template

Document Number _____
Revision _____
Revision Date _____

BUILT-IN-TEST MEASUREMENT SUMMARY FOR

SAMPLE SYSTEM/SUBSYSTEM/EQUIPMENT

Contract Number _____
Date _____

Prepared by: _____
_____

Approved by: _____
_____

Company Name
and
Address

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Sample BMS Template

Document Number _____
Revision _____
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BMS Revision Status

Date: _____
Revision: _____

Page	Revision	Page	Revision	Page	Revision	Page	Revision
i	B	41	Original				
ii	Original	42	Original				
iii	Original	43	A				
iv	Original	44	A				
v	Original	45	B				
1	Original	46	B				
2	Original	47	B				
3	Original	48	B				
4	Original	49	Original				
5	A	50	Original				
6	A						
7	A						
8	A						
9	A						
10	A						
11	B						
12	B						
13	A						
14	A						
15	B						
16	B						
17	B						
18	Original						
19	Original						
20	Original						
21	Original						
22	Original						
23	Original						
24	Original						
25	Original						
26	Original						
27	Original						
28	Original						
29	Original						
30	Original						

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Sample BMS Template

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Revision _____
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INTRODUCTION

FOR

SAMPLE
SYSTEM,
SUBSYSTEM,
EQUIPMENT

BUILT-IN-TEST MEASUREMENTS SUMMARY

[Company Name] submits a Built-In-Test Measurements Summary (BMS) in accordance with the [Sample System Name] contract Statement of Work and Contract Data Requirements List (CDRL) Item number [number] for Data Item Description (DID) for BMS.

This BMS identifies the [Sample System] measurement parameters, the Failure Modes, Effects, and Criticality Analysis (FMECA) Failure Modes associated with each measurement and test, the Built-In-Test (BIT) measurement devices, and the BIT processing components. This BMS also documents the BIT processing algorithm and details how it is used to improve the accuracy of individual pass or fail decisions. These data are required to assure measurement traceability through an unbroken chain of comparisons and ensure the adequacy of BIT to support the BIT program as specified in the contract.

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BUILT-IN-TEST MEASUREMENTS SUMMARY

TABLE OF CONTENTS

BMS Revision status i
Introduction ii
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List of Abbreviations, Symbols and Acronyms iv
Summary Data Table 1

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LIST OF ABBREVIATIONS, SYMBOLS AND ACRONYMS

TERM	DESCRIPTION
BIT	Built-In-Test
CSI	Critical Safety Item
FMECA	Failure Mode, Effects & Criticality Analysis
IBIT	Initiated Built-In-Test
ID	Identification
LRU	Line Replaceable Unit
MFR CAGE	Manufacturer Commercial and Government Entity Code
PBIT	Periodic Built-In-Test
P/N	Part Number
POST	Power On Self-Test
QUAL	Qualitative
REF DES	Reference Designation
SBIT	Startup Built-In-Test
TAR	Test Accuracy Ratio
TUR	Test Uncertainty Ratio
V	Voltage
WRA	Weapon Replaceable Assembly
Y/N	Yes/No

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SUMMARY DATA TABLE

Program:
System Model/Type Designation: AN/XXX-XXX
Date:
Version:

	Operational Equipment (System/Subsystem/WRA/LRU)		
Content ID	**Nomenclature**	**Model name**	**REF DES**
1	CONTROL,VOLTAGE REGULATOR	CN-XXXX/ALQ-XXX	A4A1
2	CONTROLLER, ELECTRICAL	MO-XX/ALQ-XXX	A5A3

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SUMMARY DATA TABLE

	Measurement		FMECA	
Content ID	Value Range or Value with Tolerance	Units	Failure Mode	CSI (Y/N)
1	28 +/- 4	V	Low Voltage	N
2	QUAL		No Communication	N

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SUMMARY DATA TABLE

	Measurement Device					
Content ID	MFR CAGE	P/N	REF DES	Range with Tolerance (accuracy)	Units	TAR
1	XXXXX	XXX	A4A1XXXX	0 – 50 +/- 1	V	4:1
2	XXXXX	XXX	A5A3XXXX			

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SUMMARY DATA TABLE

	BIT Processing Component (Level 1)			
Content ID	**MFR CAGE**	**P/N**	**REF DES**	**Comment**
1	XXXXX	XXXX	A4A1XXXX	Level 1 Component sets register XXXXX
2	N/A	N/A	N/A	Communications validated by BIT Processor

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SUMMARY DATA TABLE

	BIT Processing Component (Level 2)			
Content ID	MFR CAGE	P/N	REF DES	Comment
1	XXXXX	XXXX	A4A1XXYY	Level 2 component reports status to BIT Processor
2	N/A	N/A	N/A	

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SUMMARY DATA TABLE

	BIT Processing Algorithm			
Content ID	Test Number	Test Name	Register	Variable
1	200	CV_VOLT		V1_VAR
2	345	CE_COMM		

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SUMMARY DATA TABLE

	BIT Processing Algorithm				
Content ID	POST (Y/N)	SBIT (Y/N)	IBIT AIR (Y/N)	IBIT GROUND (Y/N)	PBIT (Y/N)
1	N	Y	Y	Y	Y
2	N	Y	N	Y	N

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SUMMARY DATA TABLE

	BIT Processing Algorithm	
Content ID	**Filtered (Y/N)**	**Processing Comments**
1	N	
2	Y	Communication attempts are attempted 3 times, 5 seconds apart. If all 3 attempts fail, then the BIT Processor logs the failure in the BIT Log file and reports the status to the host platform.

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